12TH BIOLOGY · CHAPTER 21 · FULL CHAPTER NOTES

Immunity

Topic-wise short and long questions with answers from the complete chapter

INTRODUCTION

Q.1 [Short] What is the immune system? Why are diseases like AIDS very dangerous?

The immune system is the defence system of vertebrates against infections and cancers. It continually checks the bloodstream for foreign cells and molecules, and attacks and destroys invading microbes. Diseases such as AIDS weaken the immune system, so the body cannot defend itself against infections; that is why they are very dangerous.

Q.2 [Short] Name the three lines of defence of the human body.

Line of defence	Components	Nature
First	Skin, digestive tract, respiratory tract	Non-specific
Second	Germ-killer cells of blood, protective proteins, inflammatory response, temperature response	Non-specific
Third	Immune system: monocytes, T cells, B cells	Specific

21.1 FIRST LINE OF DEFENCE

Q.3 [Long] How does the skin act as a barrier against the invasion of microbes?

- Skin is the outermost layer of the body and gives the first defence against microbes. Its epidermis is 10 to 30 cells thick.
- The outer layer of the epidermis is the **stratum corneum**. Its damaged cells are continuously shed and replaced by new cells that migrate up from the deep epidermis.
- These cells form the protein **keratin**, which makes the skin tough and impenetrable for microbes.
- The **oil and sweat glands** lower the pH at the skin surface to 3–5. This acid level inhibits the growth of many microbes.
- Sweat also contains the enzyme **lysozyme**, which attacks and digests the cell walls of many bacteria.

Q.4 [Short] How do the oil and sweat glands inhibit and kill microorganisms?

They lower the pH at the skin’s surface to 3–5; this acid level inhibits the growth of many microbes. Sweat also contains the enzyme lysozyme, which attacks and digests the cell walls of many bacteria.

Q.5 [Short] Why are the cells of the stratum corneum continuously shed?

They are continuously damaged. New cells are formed in the deep epidermis, migrate upward and replace the worn-out cells. These cells form keratin, which keeps the skin tough.

Q.6 [Short] How do the digestive and respiratory tracts defend the body?

Digestive tract: Most bacteria present in food are killed by saliva in the oral cavity, which contains lysozyme. The strong digestive acids of the stomach and the protein-digesting enzymes of the intestine also kill bacteria.

Respiratory tract: Ciliated epithelial cells in the nasal cavity trap many bacteria of the air. The cells lining the smaller bronchi and bronchioles secrete sticky mucous that traps microbes before they reach the lungs. Cilia continually sweep this mucous upward, where it is swallowed.

Q.7 [Short] What is the role of cilia and mucous in the air passages?

The sticky mucous secreted by the lining of the smaller bronchi and bronchioles traps airborne microbes before they reach the warm, moist lungs. The cilia continually sweep this mucous upward, where it can be swallowed.

21.2 SECOND LINE OF DEFENCE

Q.8 [Short] What is the second line of defence? Name its four important devices.

When invaders overcome the first line and reach deep tissues, the second line of defence acts. It is also non-specific. Its four most important devices are: germ-killer cells of blood, antimicrobial (protective) proteins, the inflammatory response and the temperature response.

Q.9 [Long] Describe the role of macrophages and neutrophils in killing bacteria.

Macrophages develop from monocytes (a type of white blood cell). A macrophage ingests a bacterium by endocytosis and kills it with its lysozyme enzymes. Some macrophages are fixed in organs such as the lungs, liver, spleen and brain, but most of them patrol the blood, lymph and interstitial fluid.

Neutrophils are a type of white blood cell. They kill pathogens by three methods:

1. **Phagocytosis:** they engulf and digest the pathogens.
2. They release soluble **anti-microbial chemicals** on the pathogens.
3. They generate and release a **network of DNA fibres**, which bind with pathogenic bacteria and kill them.

Q.10 [Short] How do natural killer cells kill infected cells and cancer cells?

Natural killer cells do not kill the invading microbes. They detect and attack the body cells that have been infected with viruses. They release special proteins called **perforins**, which insert into the membrane of the target cell and form a pore. Water rushes in through the pore, and the cell swells and bursts. They also detect cancer cells and kill them before they can develop into a tumour.

The textbook (p.173) describes natural killer cells as "a type of T-lymphocytes ... also called cytotoxic T-cells". Later (p.176–177) cytotoxic T cells are described separately as part of the specific defence, working by the same killing mechanism. Learn the textbook wording for the board exam.

Q.11 [Short] Why do the germ-killer cells not attack the body's own cells?

All normal body cells carry a surface protein marker called **self-protein**, which identifies them. The patrolling cells (macrophages, neutrophils and natural killer cells) do not attack cells that have self-proteins; they attack only the cells that lack them.

Q.12 [Long] How do complement proteins kill bacteria and interferons inhibit viruses?

The **complement system** consists of more than a dozen proteins made in the liver. They circulate in the blood in an inactive state. When they meet a microbe, they become active in a sequence: the first protein activates the second, and so on. The final five proteins form a **membrane-attack complex (MAC)** in the plasma membrane of the microbe. This complex punctures the membrane, so fluids and salts move into the microbe, and it swells and bursts.

Interferons are protective proteins released by virus-infected cells. They cause the nearby cells to enhance their anti-viral defences, so the virus cannot infect them easily. Interferons also activate other immune cells, such as natural killer cells and macrophages. They belong to the large class of proteins called **cytokines**, which work for communication between cells to trigger the protective defences of the immune system.

Q.13 [Short] What is the membrane-attack complex (MAC)?

It is the complex formed by the final five proteins of the complement system in the plasma membrane of a microbe. It punctures the membrane, so fluids and salts move in and the microbe swells and bursts.

Q.14 [Short] What are cytokines?

Cytokines are a large class of proteins that work for communication between cells, to trigger the protective defences of the immune system. Interferons belong to this class.

Q.15 [Long] State the events of the inflammatory response.

1. Infected or injured cells release chemical signals such as **histamines** and **prostaglandins**.
2. These chemicals cause local expansion of blood vessels, so the capillary blood flow increases at the site. The capillaries also become more permeable.
3. This results in **redness, swelling and warmth** at the spot. The condition is called inflammation.
4. The heat makes the site unfavourable for microbial growth. It also promotes healing and raises the mobility of white blood cells.
5. More phagocytes (macrophages and neutrophils) migrate from the blood to the inflamed tissue and engulf the microbes.

6. Monocytes arrive and release chemicals that kill the microbes; then macrophages clean up dead microbes, cells and debris.

The inflammatory response is often strong enough to stop the spread of microbes.

Q.16 [Short] Why does an inflamed spot become red, swollen and warm?

Histamines and prostaglandins expand the local blood vessels, so more blood flows to the spot (redness and warmth). The capillaries also become more permeable, so fluid leaks into the tissue (swelling).

Q.17 [Short] What is pus?

Pus is a mixture of dead or dying neutrophils, broken-down tissue cells and dead pathogens. It is associated with some infections.

Q.18 [Short] What are pyrogens? How do they cause fever?

When macrophages attack invading microbes, they release chemicals called **pyrogens**. The pyrogens travel in the blood to the **hypothalamus**, the body's thermostat, and stimulate it to raise the core temperature above the normal 37 °C (98.6 °F). The higher-than-normal temperature is called **fever** (pyrexia).

Q.19 [Short] List the ways in which fever affects microbes.

- It hinders the ability of bacteria and viruses to reproduce.
- Some temperature-sensitive microbes (e.g., *Streptococcus pneumoniae*, Influenza) are killed at high fever.
- It increases the speed of action of neutrophils and macrophages.
- During fever the body releases more cytokines (interferons).

Q.20 [Short] Which body temperatures are dangerous?

In general, temperatures greater than 103 °F (39.4 °C) are considered dangerous, and those greater than 105 °F (40.5 °C) are often fatal.

21.3 THIRD LINE OF DEFENCE

Q.21 [Short] Define the specific immune system (third line of defence).

It is the defence provided by the immune system when microbes overcome the first and second lines. It is specific: it targets particular germs and acts especially against them. It can also remember previous invaders and responds quickly if they enter again.

Q.22 [Long] Describe the types of immunity.

Type	Description
Innate (inborn) immunity	Non-specific defence (first and second lines); present at birth
Acquired immunity	Specific defence; acquired during growth; of two types, active and passive

Type	Description
Active immunity	The body itself prepares antibodies against the antigens of microbes
– Natural active	Antibodies are prepared naturally on exposure to antigens; the most effective form
– Artificial active	Weakened antigens are administered (vaccination); they stimulate antibodies against the real antigens
Passive immunity	The individual is given antibodies against specific microbes; short-lived

Q.23 [Short] Differentiate between active and passive immunity.

Active immunity	Passive immunity
The body itself prepares antibodies against the antigens of microbes.	The individual is given antibodies to combat specific microbes.
It may be natural (exposure to antigens) or artificial (vaccination).	Ready-made antibodies come from outside the body.
Memory cells are formed, so it is long-lasting.	It is short-lived.

Q.24 [Short] What is vaccination? How does a vaccine give immunity?

Vaccination is the administration of weakened antigens into the body to develop artificial active immunity. Vaccines are made from killed or weakened pathogens. They stimulate antibody production and the formation of memory cells, so the body responds rapidly if the real pathogen enters later.

Q.25 [Long] Describe monocytes, T cells and B cells as components of the immune system.

The immune system is composed of white blood cells that arise in the bone marrow and circulate in blood and lymph. They are also found in the lymph nodes, spleen, liver, thymus and bone marrow.

1. Monocytes

Monocytes are a type of WBC. They circulate in the blood and enter the interstitial fluid, where they mature into **macrophages**. Macrophages phagocytose and destroy microbes, and produce proteins that activate helper T cells. Monocytes also mature into other phagocytes that stay in the liver, spleen, lymph nodes and other tissues. They help to clear cellular debris and repair tissue damage.

2. T cells

T cells develop from stem cells in the bone marrow and migrate to the **thymus** (a small grey gland just above the heart). There they mature and gain the ability to identify the antigens of invading microbes. Tens of millions of different T cells are made, each specialized for one particular antigen. There are four main kinds:

Kind of T cell	Role
Helper T cells	Initiate the immune response
Cytotoxic T cells	Break the cells that have been infected by microbes
Inducer T cells	Supervise the development of T cells in the thymus
Suppressor T cells	Terminate the immune response

3. B cells

B cells also develop in the bone marrow, but they do not travel to the thymus; they complete their maturation in the bone marrow and are then released into the blood and lymph. Each B cell is specialized to recognize a particular antigen. When a B cell meets its antigen, it divides into many **plasma cells**. Each plasma cell produces an antibody that sticks to the antigen and marks it for destruction.

Q.26 [Short] Where do T cells and B cells mature?

Both develop from stem cells in the bone marrow. T cells migrate to the thymus and mature there. B cells do not travel to the thymus; they complete their maturation in the bone marrow.

Q.27 [Short] What are plasma cells?

They are the many new cells formed when a B cell meets its antigen and divides. Each plasma cell produces an antibody that sticks to the antigen and marks it for destruction.

Q.28 [Short] How are helper T cells activated?

When a macrophage engulfs a microbe, it produces special proteins for the helper T cells. The helper T cells recognize these proteins through their receptors and become activated. The activated helper T cells start two parallel immune responses: cell-mediated immunity and antibody-mediated immunity.

Q.29 [Long] Describe the role of T cells in cell-mediated immunity.

Cell-mediated immunity is carried out by T cells. It protects the body from virus infections and cancer by killing abnormal or virus-infected cells. Its events are:

1. **Proliferation:** The activated helper T cell produces soluble chemicals, e.g., T cell growth factor. It starts the proliferation of all types of T cells, so large clones of T cells that recognize the antigen are formed.
2. **Activation:** A second factor from the helper T cell attracts macrophages to the site of infection.
3. **Induction:** Helper T cells activate inducer T cells in the thymus, which trigger the maturation of lymphocytes into T cells.
4. **Attack:** Receptors on the cytotoxic T cells recognize the infected body cells. The cytotoxic T cells bind with them and disrupt their membranes, so the cells burst.
5. **Suppression:** Suppressor T cells multiply more slowly than cytotoxic T cells. After 1 to 2 weeks their number rises enough to shut down the cytotoxic T cell response.

6. **Memory:** A population of helper and cytotoxic T cells persists, probably for life, as **memory cells**. They give a rapid response if the same microbe enters again.

Q.30 [Short] Why can suppressor T cells not stop the cytotoxic attack at once?

Suppressor T cells multiply more slowly than cytotoxic T cells. Their low initial number cannot block the attack. After 1 to 2 weeks their number rises enough to shut down the cytotoxic T cell response.

Q.31 [Long] Describe the role of B cells in antibody-mediated immunity.

Antibody-mediated (humoral) immunity is a second, stronger defence activated by the helper T cells. Its key players are the B cells. They do not attack antigens directly; they mark them for destruction and also inactivate toxic molecules. Its events are:

1. **Proliferation:** Each B cell binds to the antigen through the B receptor proteins on its surface. Helper T cells recognize such antigen-bound B cells, bind with them and release proteins that stimulate the B cell to proliferate. A large number of **plasma cells** are produced.
2. **Differentiation and secretion:** Most plasma cells stop reproducing and start making more B receptor proteins. When secreted, these proteins are called **antibodies**. A great number of antibodies are secreted in only a few days.
3. **Attack:** Antibodies mark the antigen for destruction in three ways. (i) *Complement:* complement proteins make holes in the antibody-coated antigen cell, which swells and bursts. (ii) *Macrophages:* they recognize the antibody-coated antigen, ingest it and break it. (iii) *Killer cells:* they have receptors for antibody-coated antigens; they bind and kill the cell.
4. **Suppression:** After several weeks, suppressor T cells shut down the antibody response.
5. **Memory:** Some plasma cells do not produce antibodies; they persist as **memory B cells**. They give a rapid secondary response if the same antigen enters again.

Q.32 [Short] By which three mechanisms do antibodies mark an antigen for destruction?

(1) **Complement:** the attached antibodies activate complement proteins, which make holes in the antigen cell. (2) **Macrophages:** they recognize the antibody-coated antigen and ingest it. (3) **Killer cells:** they have receptors for antibody-coated antigens; they bind and kill the cell.

Q.33 [Short] Explain the role of memory cells in long-term immunity.

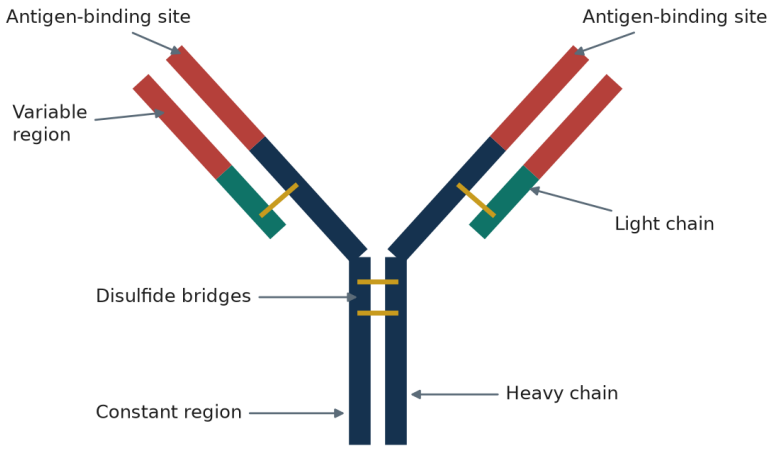
After the immune response is suppressed, some helper and cytotoxic T cells and some B cells persist as memory cells, probably for the life of the individual. If the same microbe or antigen enters again, they give a rapid secondary response, so the infection is controlled quickly.

Q.34 [Short] How may gene therapy help the immune system against malignant melanoma?

Malignant melanoma is a lethal skin cancer. Tumour-infiltrating lymphocytes (TIL) attack the tumour but are not strong enough. Genetic engineers are trying to insert a gene into TIL so that they make tumour necrosis factor (TNF), a protein that kills tumour cells. The engineered TIL, returned to the patient's blood, would enter the tumour and kill it.

Q.35 [Long] Describe the structure of an antibody.

An antibody is a **Y-shaped protein**. It is composed of four polypeptide chains: two identical long (**heavy**) chains and two short (**light**) chains, held together by **disulfide bridges**. Each chain has a **constant region** (similar in all antibodies of the same class) and a **variable region** (differs among antibodies). The variable regions of one heavy and one light chain on each arm of the Y form a specific **binding site**. Because different antibodies have different variable regions, each antibody is specific for the antigen that stimulated its production.



Structure of an antibody

Q.36 [Short] Why are antibodies specific?

The binding site of an antibody is formed by the variable regions of one heavy and one light chain. Different antibodies have different variable regions. So each antibody binds only the specific antigen that stimulated its production.

Q.37 [Long] What are monoclonal antibodies? Describe their discovery and importance.

Only a small portion of an antigen, called the **determinant**, fits into the binding site of an antibody. Antibodies that attach to different portions of the same antigen are **polyclonal antibodies**. Antibodies that attach with only one determinant are **monoclonal antibodies**.

Discovery: Cesar Milstein and George Kohler received the Nobel Prize in 1984 for learning how to produce a monoclonal antibody. They mixed plasma cells (which produce antibodies) with cancer cells called **myelomas**. New cells called **hybridomas** grew in the culture. Hybridomas used the genes of the plasma cell to make the antibody

and the myeloma genes to grow and divide. So they obtained rapidly growing cells that kept producing one monoclonal antibody.

Importance

- They are used to purify specific molecules from complex mixtures; interferon was purified in this way.
- In medicine, they are used as vehicles for delivering specific therapies.
- Research is under way to target radioactive monoclonal antibodies against antigens found only on cancer cells, to kill cancer cells selectively.
- They are used to destroy the T cells that attack a transplant, to prevent transplant rejection.

Q.38 [Short] Differentiate between polyclonal and monoclonal antibodies.

When many different antibodies attach to one antigen, each fitting a different portion of its surface, they are **polyclonal antibodies**. Antibodies that can attach with only one determinant are **monoclonal antibodies**.

Q.39 [Short] What are hybridomas?

They are the cells formed when plasma cells are mixed with cancer cells called myelomas. They use the plasma-cell genes to make one antibody and the myeloma genes to grow and divide rapidly.

21.3.3 DISORDERS OF IMMUNE SYSTEM

Q.40 [Long] Define allergies. Correlate their symptoms with the release of histamines.

Allergies result from the hypersensitivity of the immune system to weak antigens that do not cause an immune response in most people. The substances that cause allergies are called **allergens**, e.g., dust, molds, pollen, cat dander, certain foods and some medicines (such as penicillin).

1. After exposure to an allergen, some people make a kind of antibody called **E-antibodies**.
2. These antibodies bind to **mast cells** (found in connective tissue around blood vessels).
3. The mast cells release **histamines**, which start the inflammatory response.
4. Histamines cause dilation of blood vessels and leakage of fluid. This produces the symptoms: sneezing, runny nose, itching etc.
5. In some individuals, histamines cause life-threatening **anaphylactic shock**, in which swelling makes breathing difficult.

Treatment: Antihistamine drugs block histamines and give temporary relief. In **desensitization**, extracts of the antigens are injected into the patient. This produces G-antibodies in the blood, which deal with the allergen before it meets the E-antibodies.

Q.41 [Short] What is desensitization?

It is a treatment of allergies in which extracts of the antigens are injected into the patient. It produces G-antibodies in the blood, which deal with the allergen before it meets the E-antibodies.

Q.42 [Short] What is anaphylactic shock?

It is a life-threatening allergic reaction caused by histamines in some individuals, in which swelling makes breathing difficult.

Q.43 [Long] Describe autoimmune diseases with examples.

Autoimmune diseases result when the immune system is unable to distinguish between body cells and foreign materials. The immune system then attacks and destroys the cells and tissues of its own body. Examples:

- **Myasthenia gravis:** antibodies against the acetylcholine receptors of the patient's own skeletal muscle cells; the muscles do not respond to acetylcholine, causing muscle weakness.
- **Systemic lupus erythematosis (SLE):** antibodies against the patient's own connective tissues and major organs.
- **Rheumatoid arthritis:** antibodies against the cartilaginous tissue of joints, which is damaged.
- **Type I (juvenile) diabetes:** antibodies destroy the insulin-producing cells of the pancreas.

Most medicines for autoimmune diseases suppress immunity in general.

Q.44 [Short] Why is inflammation in arthritis called a misdirected immune response?

In arthritis, inflammation occurs in the absence of infection or injury. In rheumatoid arthritis, antibodies are formed against the cartilage of the patient's own joints. So the response is directed against the body instead of microbes.

Q.45 [Long] Discuss the role of T cells and B cells in transplant rejection.

Normally, T and B cells attack only foreign molecules and cells, because every person's cells carry a particular set of **self-proteins**, like molecular "fingerprints", which the person's own T and B cells recognize.

When a person receives an organ transplant, the self-proteins of the donated organ may not match those of the recipient. Then the recipient's **T cells attack the cells** of the transplanted organ and the **B cells produce antibodies** against its proteins. This is called **transplant rejection**.

How rejection is minimized

- Doctors look for a donor whose self-proteins match the recipient's as closely as possible. Preference order: identical twin > sibling > parent > unrelated donor.
- Drugs are used to suppress the recipient's immune system; but they also reduce the ability to fight infections.
- Monoclonal antibodies are used to destroy the T cells that attack the transplant.

- Stem cells are used to make a new immune system that recognizes the transplant as self.

Q.46 [Short] Which donor gives the best match for a transplant?

Identical twins have a 100% match of self-proteins. The best matches occur within a family: identical twin > sibling > parent > unrelated donor. The chance of an unrelated donor matching is 1 in 100,000–200,000.

In the textbook, "Transplant Rejection" is numbered "2-" (the same number as Autoimmune Diseases); it is the third disorder.

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